

CubeSat Anomaly Detection

Lightweight Few-Shot Self-Distilled Framework — Technical Report

Metric	Value
Teacher PA-F1 (5 seeds)	0.8702 ± 0.0198
Student PA-F1 (5 seeds)	0.8609 ± 0.0128
Student Model Size	1.64 KB (< 2 KB CubeSat RAM budget ✓)
Size Reduction vs Teacher	71.7% (5.79 KB → 1.64 KB)
Primary Dataset	NASA SMAP/MSL — 81 channels, 25 features
Cross-Mission Datasets	ESA OPS-SAT-1 + ESA-ADB
Random Seeds	5 seeds: 42, 7, 123, 2024, 99
Report Date	August 14, 2026

Abstract. A 9-phase pipeline pre-trains a ConvAE Teacher on unlabeled NASA SMAP/MSL telemetry via masked reconstruction, then distills it to a 1.64 KB TinyConvAE Student. MAML meta-learning provides a few-shot baseline. The student achieves 86.09% point-adjusted F1 (5 seeds) while meeting strict CubeSat RAM constraints and demonstrating cross-mission transfer to ESA OPS-SAT and ESA-ADB telemetry.

1. System Overview — 9-Phase Pipeline

Phase	Name	Key Operation	Output File
1	Baseline Training	Train LSTM, ConvAE, TransformerAE, ELM on SMAP	baseline_comparison_with_maml.csv
2	Pre-trained Self-Supervised	Self-supervised masked reconstruction pre-training + K-shot finetuning	pretrained_fewshot_curve.csv
3	MAML Meta-Learning	150 meta-epochs, inner-LR=0.001, 3 inner steps, 8 tasks/batch	maml_fewshot_curve.csv
4	Knowledge Distillation	Teacher ConvAE → TinyConvAE Student, 30 epochs, alpha=0.01	distilled_student_fewshot_curve.csv
5	ESA Cross-Mission	Zero-shot + K-shot transfer to ESA-ADB telemetry	esa_adb_transfer_results.csv
6	Power/EPS Case Study	Isolate P-prefix channels, 5-fold cross-validation	power_subsystem_breakdown_kfold.csv
7	Cost-Sensitive Asymmetric	ReconLoss — EXCLUDED from core results (invalid)	EXCLUDED
8	Health Index	Weighted per-channel scores → single scalar (0-100)	health_index_breakdown.csv
9	Multi-Seed Reproducibility	5 runs: 42, 7, 123, 2024, 99 — mean ± std for all models	multiseed_results.csv

2. Libraries & Software Environment

Compute Environment

Component	Specification
Runtime	Google Colab — NVIDIA T4 GPU (16 GB) or CPU fallback
Storage	Google Drive — /content/drive/MyDrive/cubesat_project
Python	Python 3.10+ (Colab default)
CUDA	Auto-detected via torch.cuda.is_available()
Notebook	CubeSat_Anomaly_Detection_13_8_v7_ALL_CHECKPOINTS.ipynb

Python Library Stack

Library	Version	Role
torch (PyTorch)	≥ 2.0	All neural networks, MAML (torch.func.functional_call), checkpointing
torch.nn	built-in	ConvAE, TinyConvAE, LSTMForecaster, TransformerAE
torch.optim.Adam	built-in	Optimizer for all models
torch.func	≥ 2.0	MAML second-order gradient computation
torch.nn.utils.prune	built-in	Structural pruning on student model
numpy	≥ 1.23	Array ops, sliding windows, normalization
pandas	≥ 1.5	CSV I/O, labeled_anomalies.csv parsing
scikit-learn	≥ 1.2	precision/recall/f1_score, ELM random projections
matplotlib	≥ 3.6	All 300 DPI publication figures
tqdm	≥ 4.65	Progress bars for channel iteration and training
os, json, shutil	stdlib	Path management, checkpoint I/O, file migration
urllib.request	stdlib	Automated NASA dataset download
ast	stdlib	Safe parsing of anomaly_sequences column
google.colab.drive	Colab	Google Drive mounting

3. Dataset Description

3.1 Primary Dataset — NASA SMAP/MSL

Property	Details
Source	NASA JPL (public) — auto-downloaded via urllib.request
URL	https://s3-us-west-2.amazonaws.com/teleanom/data.zip
Channels	81 telemetry channels (55 SMAP + 26 MSL)
Features/step	25 features per timestep per channel
Train length	~2,000–8,000 timesteps per channel
Test length	~500–4,000 timesteps per channel
Anomaly rate	~12–27% of test timesteps per channel
Labels	labeled_anomalies.csv — chan_id, anomaly_sequences [[start,end],...]
Train/Test split	Pre-split: data/train/*.npy and data/test/*.npy
Normalization	Per-channel z-score: mean/std from TRAIN only, applied to both splits

3.2 ESA Cross-Mission Datasets

Dataset	Source	Mission	Use in Project
OPS-SAT-AD	ESA / Kaggle	ESA OPS-SAT-1 (2020–now)	Phase 3: zero-shot + K-shot (3,5,10,20) evaluation
ESA-ADB	GitHub esa/anomaly-dataset	Multi-mission ESA archive	ESA_ADB_Cross_Mission_Extension.ipynb evaluation

4. Data Preprocessing Pipeline

Step	Function	Parameters	Description
1. Load	load_channel(chan_id, split)	—	np.load(data/{split}/{chan_id}.npy)
2. Normalize	normalize(train, test)	$\epsilon = 1e-8$	$z = (x - \mu_{\text{train}})/(\sigma_{\text{train}} + \epsilon)$. No test leakage.
3. Windows	make_windows(arr)	WINDOW=100, STRIDE=10	Sliding windows $\rightarrow (n_{\text{windows}}, 100, n_{\text{features}})$
4. Labels	label_windows(starts, window)	Overlap-based	A window is anomalous if ANY timestep overlaps an anomaly segment
5. MAML norm	normalize_channel_for_maml()	MIN_STD=0.05	Per-channel MAML norm; clips σ at 0.05 to prevent blow-up
6. Threshold	Dynamic per-channel 99th pct	np.percentile(scores,99)	One threshold per channel computed on test scores
7. Point-Adj	Point-adjust inside channel	Per-channel boundary	Flag entire anomaly segment if ANY window detected. Standard SMAP/MSL protocol

5. Model Architectures

5.1 ConvAE — Teacher Backbone

Layer	Type	In Ch	Out Ch	Kernel	Padding	Activation
Encoder 1	Conv1d	n_feat	16	5	2	ReLU
Encoder 2	Conv1d	16	8	5	2	ReLU
Decoder 1	Conv1d	8	16	5	2	ReLU
Decoder 2	Conv1d	16	n_feat	5	2	Linear

Parameters: 1,482 | Size: 5.79 KB (float32) | Input: (batch, 100, n_features)

5.2 TinyConvAE — Distilled Student (onboard deployment model)

Layer	Type	In Ch	Out Ch	Kernel	Padding	Activation
Encoder 1	Conv1d	n_feat	8	5	2	ReLU
Encoder 2	Conv1d	8	4	5	2	ReLU
Decoder 1	Conv1d	4	8	5	2	ReLU
Decoder 2	Conv1d	8	n_feat	5	2	Linear

Parameters: 420 | Size: 1.64 KB (float32) | 71.7% smaller than Teacher | Meets <2 KB CubeSat RAM budget ✓

5.3 Other Baseline Models

Model	Architecture	Params	Size	Mode
LSTMForecaster	2-layer LSTM (hidden=64) + Linear head	~158,000	~618 KB	Forecasting
TransformerAE	Linear(n→32) + 2-layer TransformerEncoder + Linear(32→n)	~61,000	~241 KB	Reconstruction
ELM_AE	Random hidden layer (300 units) + pseudoinverse output	~60,000	~235 KB	Closed-form
ConvAE+Dropout	ConvAE with Dropout(p=0.15), T=30 MC passes	~1,500	~5.79 KB	Uncertainty

6. Training Configurations & Hyperparameters

Model / Phase	Epochs	Batch	LR	Optimizer	Loss	Checkpoint File
ConvAE (baseline)	30	64	1e-3	Adam	MSELoss	main_ConvAE_latest.pth
LSTMForecaster	30	64	1e-3	Adam	MSELoss (last timestep)	main_LSTM_latest.pth
TransformerAE	30	64	1e-3	Adam	MSELoss	main_Transformer_latest.pth
ELM_AE	1	N/A	N/A	Pseudoinv	MSE (closed-form)	N/A
ConvAE Pretrain (Teacher)	30	64	1e-3	Adam	MSELoss (masked reconstruction)	main_PretrainedFewShot_latest.pth
MAML (meta-train)	15	8 tasks	outer 1e-3 inner 0.001	Adam	MSELoss (query set)	main_MAML_latest.pth
Student Distillation	30	64	1e-3	Adam	$0.7 \cdot \text{MSE}(s,x) + 0.3 \cdot \text{MSE}(s,t)$	main_student_latest.pth
Few-shot fine-tune	50 steps	support	1e-3	Adam	MSELoss	In-memory
Multi-seed reruns	30	64	1e-3	Adam	MSELoss	seed-specific .pth

MAML Hyperparameters

Parameter	Value	Description
K_SHOT	5	Support set size per task
Q_SIZE	15	Query set size per task
INNER_LR	0.001	Inner-loop learning rate
INNER_STEPS	3	Gradient steps per inner loop
META_EPOCHS	15	Outer-loop training epochs
TASKS_PER_BATCH	8	Channel-tasks sampled per meta-update
Train/Test split	80/20%	Random channel split for meta-evaluation

7. Checkpointing System

All epoch-based models use `train_torch_model_checkpoint_unified()` which saves after every epoch into a unified .pth file. Format: {model_state, optimizer_state, epoch, loss}. Colab can resume training after timeout without losing progress.

File	Model	Size	Purpose
main_PretrainedFewShot_latest.pth	ConvAE Teacher	5.79 KB	Core teacher — main result model
main_student_latest.pth	TinyConvAE Student	1.64 KB	★ Distilled student — deploy this onboard
main_MAML_latest.pth	ConvAE MAML	5.79 KB	MAML few-shot baseline
main_ConvAE_latest.pth	ConvAE baseline	5.79 KB	Phase 1 standard baseline
main_LSTM_latest.pth	LSTMForecaster	~618 KB	LSTM baseline
main_Transformer_latest.pth	TransformerAE	~241 KB	Transformer baseline
main_MCDropout_latest.pth	ConvAE+Dropout	~5.79 KB	Uncertainty estimation
main_*_seed{N}_latest.pth × 30	6 models × 5 seeds	varies	Multi-seed reproducibility checkpoints

8. Experimental Results

All numbers from multiseed_results.csv — source of truth

8.1 Multi-Seed Headline Results (5 Seeds)

Rank	Model	PA-F1 Mean	PA-F1 Std	Seeds	Model Size
1	Pretrained+FewShot (5-shot)	0.8702	0.0198	5	5.79 KB
2	Distilled Student (5-shot)	0.8609	0.0128	5	1.64 KB ★
3	MAML (5-shot)	0.8121	0.0856	5	5.79 KB
4	ELM	0.7771	0.0333	5	~235 KB
5	Transformer	0.7240	0.0454	5	~241 KB
6	Conv-AE (1D)	0.7138	0.0392	5	5.79 KB
7	LSTM Forecaster	0.6595	0.0310	5	~618 KB

8.2 Per-Seed Breakdown — Teacher & Student

Model	Seed	Raw F1	PA Precision	PA Recall	PA F1
Distilled Student (5-shot)	7	0.1410	0.9594	0.7956	0.8698
Distilled Student (5-shot)	42	0.1303	0.9405	0.8103	0.8705
Distilled Student (5-shot)	99	0.0947	0.9611	0.7956	0.8705
Distilled Student (5-shot)	123	0.0911	0.9298	0.7766	0.8463
Distilled Student (5-shot)	2024	0.1033	0.9254	0.7817	0.8475
Pretrained+FewShot (5-shot)	7	0.1253	0.9893	0.8117	0.8918
Pretrained+FewShot (5-shot)	42	0.1100	0.9307	0.7670	0.8410
Pretrained+FewShot (5-shot)	99	0.1008	0.9900	0.7956	0.8822
Pretrained+FewShot (5-shot)	123	0.0799	0.9748	0.7941	0.8753
Pretrained+FewShot (5-shot)	2024	0.0967	0.9378	0.7956	0.8609

8.3 ESA-ADB Cross-Mission Transfer

Model	K-Shot	Raw F1	Point-Adjust F1	Model Size (KB)
Teacher (Zero-Shot)	0.0000	0.3215	0.6039	5.7900
Distilled Student (Zero-Shot)	0.0000	0.3359	0.6075	1.6400
Teacher (3-Shot)	3.0000	0.3635	0.6974	5.7900
Distilled Student (3-Shot)	3.0000	0.1837	0.6252	1.6400
Teacher (5-Shot)	5.0000	0.1322	0.6980	5.7900
Distilled Student (5-Shot)	5.0000	0.0896	0.6288	1.6400
Teacher (10-Shot)	10.0000	0.0746	0.6842	5.7900
Distilled Student (10-Shot)	10.0000	0.1067	0.6689	1.6400

Model	K-Shot	Raw F1	Point-Adjust F1	Model Size (KB)
Teacher (20-Shot)	20.0000	0.0626	0.6586	5.7900
Distilled Student (20-Shot)	20.0000	0.0751	0.6614	1.6400

8.4 OPS-SAT Zero-Shot Transfer

Model	Raw F1	PA Precision	PA Recall	PA F1
Teacher (zero-shot)	0.0938	0.6538	0.1504	0.2446
Distilled Student (zero-shot)	0.0763	0.5357	0.1327	0.2128

9. Key Mathematical Formulations

Name	Formula
Z-Score Normalization	$z(t) = (x(t) - \text{mean_train}) / (\text{std_train} + 1e-8)$
Reconstruction Loss (Train)	$L = (1/B \times T \times F) * \sum((x - \hat{x})^2)$
Distillation Loss	$L = 0.7 * \text{MSE}(\text{student_out}, x) + 0.3 * \text{MSE}(\text{student_out}, \text{teacher_out})$
Anomaly Score (per window)	$\text{score}(w) = (1/T \times F) * \sum((x_w - \hat{x}_w)^2)$
Dynamic Threshold	$\tau_c = 99\text{th-percentile of test scores for channel } c$
Point-Adjusted Prediction	$y_{PA}(w) = 1 \text{ if } \text{score}(w) > \tau_c \text{ OR } w \text{ belongs to detected segment}$
MAML Inner Update	$\theta'_i = \theta - \alpha_{\text{inner}} * \text{grad}_{\theta}(L_{\text{task}_i})$
MAML Outer Update	$\theta = \theta - \alpha_{\text{outer}} * \text{grad}_{\theta}(\sum_i L(f_{\theta'_i}))$
Health Index	$HI(t) = 100 * (1 - \sum_i (w_i * \text{score}_i(t)) / \sum_i (w_i))$
Compression Ratio	$CR = 1 - (420 / 1482) = 71.7\%$

10. Known Limitations

Must be disclosed in paper

#	Limitation	Severity	Required Paper Action
1	few-shot curve is flat (F1 range < 0.002 across K=3-20). More labels are added	MEDIUM	Explicitly state this is a limitation. Do NOT present as data-efficiency result.
2	ADB mean F1 (0.253) much higher than OPS-SAT (0.172). May reflect differences in data	MEDIUM	Clarify test conditions. Do not use identical point-adjust window logic before citing numbers
3	Power8-channel K-fold F1 = 0.0 in 4 of 5 folds due to tiny sample size	LOW	Mark power case study EXPLORATORY ONLY. Do not headline.
4	Cost-sensitive loss experiment invalid — training used unlabeled data, reducing	HIGH	Do NOT report cost-weighted F1 anywhere in the paper.
5	Power seeds is underpowered for Wilcoxon test. Teacher vs Student gap is not statistically significant	MEDIUM	Do not claim 'statistically significant' difference only. Do not claim statistical significance.

11. Project File Structure

Google Drive Path: /content/drive/MyDrive/cubesat_project

```
cubesat_project/  
CubeSat_Anomaly_Detection_13_8_v7_ALL_CHECKPOINTS.ipynb [MAIN – 618 KB, 134 cells]  
CubeSat_Results_Analysis.ipynb [Paper figures & LaTeX tables]  
ESA_ADB_Cross_Mission_Extension.ipynb [ESA-ADB standalone extension]  
data/  
labeled_anomalies.csv [Ground truth labels]  
train/{chan_id}.npy x 81 [Training telemetry]  
test/{chan_id}.npy x 81 [Test telemetry]  
checkpoints/ [38 .pth files – KEEP ALL]  
main_PretrainedFewShot_latest.pth 5.79 KB Teacher backbone  
main_student_latest.pth 1.64 KB Student – deploy this  
main_MAML_latest.pth 5.79 KB MAML model  
main*_seed{N}_latest.pth x 30 varies Multi-seed reproducibility  
results/tables/ [22 CSV result files]  
multiseed_results.csv SOURCE OF TRUTH – 35 rows x 9 columns  
headline_summary.csv Rebuilt from multiseed_results  
paper_latex_headline_table.tex Ready for LaTeX paper insertion  
LIMITATIONS_NOTES.txt Auto-generated caveats checklist  
FINAL_SUBMISSION_20260814.zip Complete results archive  
results/figures/ [8 PNG files, 300 DPI]  
paper_fig1_pipeline_progression.png  
paper_fig2_fewshot_data_efficiency.png  
paper_fig3_power_subsystem_breakdown.png  
paper_fig4_opssat_cross_mission_transfer.png  
esa_adb_cross_mission_transfer.png  
CubeSat_Checkpoints-2/ [SAFE TO DELETE – legacy]
```